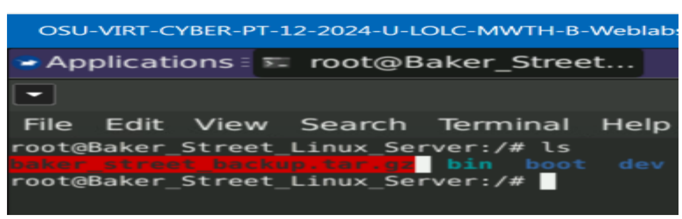
Cybersecurity

Hardening Summary and Checklist

OS Information

Customer	Baker Street Corporation
Hostname	**Baker_Street_Linux_Server**
OS Version	**Version="22.04.5 LTS (Jammy Jellyfish)"**
Memory information	**total used free shared buff/cache available** **Mem: 15Gi 1.4Gi 7.3Gi 184Mi 6.7Gi 13Gi** **Swap: 0B 0B 0B**
Uptime information	**Up 27 minutes**

Checklist

Completed	Activity	Script(s) used / Tasks completed / Screenshots
☑	OS backup	

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<td>☑</td>
<td>Validating and updating permissions on files and directories</td>
<td>
To validate and update permissions on files and directories the following steps were taken:

This command finds files that have world permissions and removes them:
find /home -type f -perm /o+rwx -exec chmod o-rwx {} +

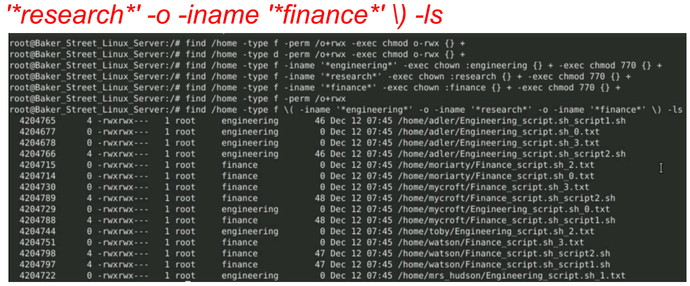
This command finds and ensures directories in home folders have no world access:
find /home -type d -perm /o+rwx -exec chmod o-rwx {} +

The command that allows only engineering group to have access:
find /home -type f -iname '*engineering*' -exec chown :engineering {} + -exec chmod 770 {} +

The command that allows only research group to have access:
find /home -type f -iname '*research*' -exec chown :research {} + -exec chmod 770 {} +

The command that allows only finance group to have access:
find /home -type f -iname '*finance*' -exec chown :finance {} + -exec chmod 770 {} +

This command checks if any files still have world permissions:
find /home -type f -perm /o+rwx

Lastly this command will list the updated files so you can verify the changes:
find /home -type f \(-iname '*engineering*' -o -iname '*research*' -o -iname '*finance*' \) -ls


I did not find any hidden files containing passwords after I ran the command:
find /home -type f -name ".*" -exec grep -Ei 'password|passwd|pwd|secret|key|token' {} + 2>/dev/null


However, after doing a search on the home directory hidden files using the command find /home
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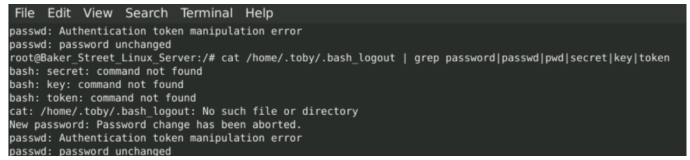
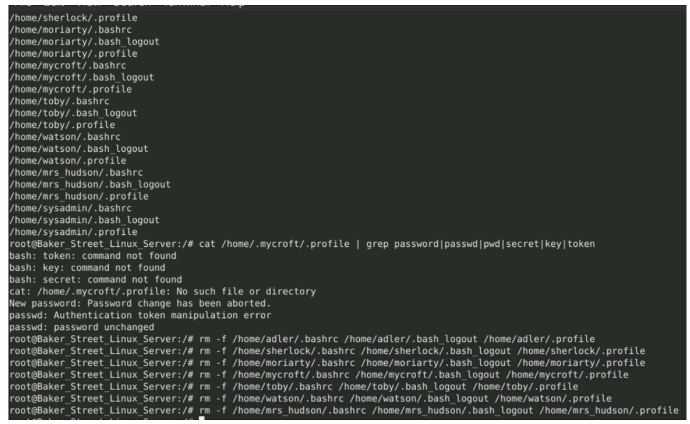
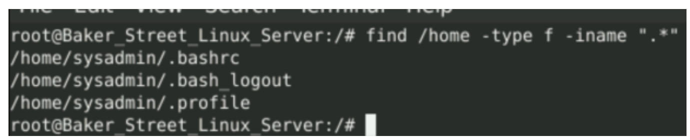
<td>-type f -iname ".*". I discovered several employees had hidden files containing passwords. I removed the files by the command:
rm -f /path/to/hidden/file

File Edit View Search Terminal Help
passwd: Authentication token manipulation error
passwd: password unchanged
root@Baker_Street_Linux_Server:/# cat /home/.toby/.bash_logout | grep password|passwd|pwd|secret|key|token
bash: secret: command not found
bash: key: command not found
bash: token: command not found
cat: /home/.toby/.bash_logout: No such file or directory
New password: Password change has been aborted.
passwd: Authentication token manipulation error
passwd: password unchanged

/home/sherlock/.profile
/home/moriarty/.bashrc
/home/moriarty/.bash_logout
/home/moriarty/.profile
/home/mycroft/.bashrc
/home/mycroft/.bash_logout
/home/mycroft/.profile
/home/toby/.bashrc
/home/toby/.bash_logout
/home/toby/.profile
/home/watson/.bashrc
/home/watson/.bash_logout
/home/watson/.profile
/home/mrs_hudson/.bashrc
/home/mrs_hudson/.bash_logout
/home/mrs_hudson/.profile
/home/sysadmin/.bashrc
/home/sysadmin/.bash_logout
/home/sysadmin/.profile
root@Baker_Street_Linux_Server:/# cat /home/.mycroft/.profile | grep password|passwd|pwd|secret|key|token
bash: token: command not found
bash: key: command not found
bash: secret: command not found
cat: /home/.mycroft/.profile: No such file or directory
New password: Password change has been aborted.
passwd: Authentication token manipulation error
passwd: password unchanged
root@Baker_Street_Linux_Server:/# rm -f /home/adler/.bashrc /home/adler/.bash_logout /home/adler/.profile
root@Baker_Street_Linux_Server:/# rm -f /home/sherlock/.bashrc /home/sherlock/.bash_logout /home/sherlock/.profile
root@Baker_Street_Linux_Server:/# rm -f /home/moriarty/.bashrc /home/moriarty/.bash_logout /home/moriarty/.profile
root@Baker_Street_Linux_Server:/# rm -f /home/mycroft/.bashrc /home/mycroft/.bash_logout /home/mycroft/.profile
root@Baker_Street_Linux_Server:/# rm -f /home/toby/.bashrc /home/toby/.bash_logout /home/toby/.profile
root@Baker_Street_Linux_Server:/# rm -f /home/watson/.bashrc /home/watson/.bash_logout /home/watson/.profile
root@Baker_Street_Linux_Server:/# rm -f /home/mrs_hudson/.bashrc /home/mrs_hudson/.bash_logout /home/mrs_hudson/.profile

root@Baker_Street_Linux_Server:/# find /home -type f -iname ".*"
/home/sysadmin/.bashrc
/home/sysadmin/.bash_logout
/home/sysadmin/.profile
root@Baker_Street_Linux_Server:/#</td>
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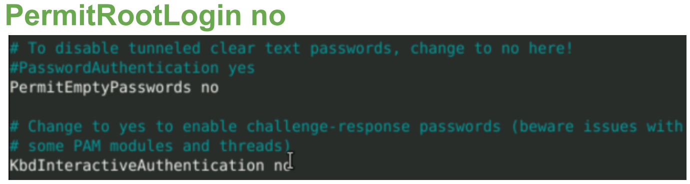
<td>☑</td>
<td>Auditing and securing SSH</td>
<td>Started with the command sudo nano /etc/ssh/sshd_config to start making edits.

PermitEmptyPasswords yes, what changed to PermitEmptyPasswrds no

PermitRootLogin yes, what changed to PermitRootLogin no

To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
PermitEmptyPasswords no

Change to yes to enable challenge-response passwords (beware issues with
some PAM modules and threads)
KbdInteractiveAuthentication no

Removed all other Ports, then uncommented the line that contained Port 22. SSH with any other ports besides 22

The strategy used for options in the default sshd_config shipped with
OpenSSH is to specify options with their default value where
possible, but leave them commented. Uncommented options override the
default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 22</td>
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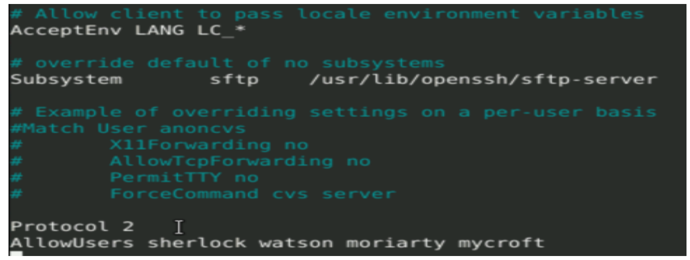
<td>Removed all lines containing different Protocols beside Protocol 2


Allow client to pass locale environment variables
AcceptEnv LANG LC_*

override default of no subsystems
Subsystem sftp /usr/lib/openssh/sftp-server

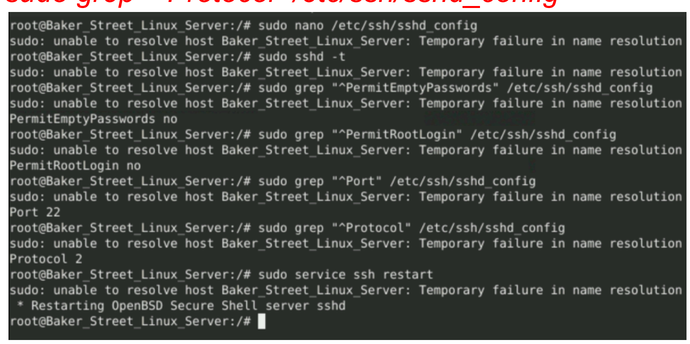
Example of overriding settings on a per-user basis
#Match User anoncvs
X11Forwarding no
AllowTcpForwarding no
PermitTTY no
ForceCommand cvs server

Protocol 2
AllowUsers sherlock watson moriarty mycroft
Verified changes were correct the running the following commands:
sudo sshd -t

sudo grep "^PermitEmptyPasswords" /etc/ssh/sshd_config

sudo grep "^PermitRootLogin" /etc/ssh/sshd_config

sudo grep "^Port" /etc/ssh/sshd_config

sudo grep "^Protocol" /etc/ssh/sshd_config

root@Baker_Street_Linux_Server:/# sudo nano /etc/ssh/sshd_config
sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in name resolution
root@Baker_Street_Linux_Server:/# sudo sshd -t
sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in name resolution
root@Baker_Street_Linux_Server:/# sudo grep "^PermitEmptyPasswords" /etc/ssh/sshd_config
sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in name resolution
PermitEmptyPasswords no
root@Baker_Street_Linux_Server:/# sudo grep "^PermitRootLogin" /etc/ssh/sshd_config
sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in name resolution
PermitRootLogin no
root@Baker_Street_Linux_Server:/# sudo grep "^Port" /etc/ssh/sshd_config
sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in name resolution
Port 22
root@Baker_Street_Linux_Server:/# sudo grep "^Protocol" /etc/ssh/sshd_config
sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in name resolution
Protocol 2
root@Baker_Street_Linux_Server:/# sudo service ssh restart
sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in name resolution
* Restarting OpenBSD Secure Shell server sshd
root@Baker_Street_Linux_Server:/#

Then applied changes by running the command:
sudo service ssh restart</td>
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<td>☑</td>
<td>Reviewing and updating system packages</td>
<td>To Review and update system packages. I first ran the command sudo apt update to ensure I was working with the latest version of all packages.

Next I ran sudo apt upgrade -y to update all installed packages to their latest versions.

To view installed packages I ran the command sudo apt list –installed. Also created the file package_list.txt.</td>
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```
root@Baker_Street_Linux_Server:/# sudo apt update
sudo: unable to resolve host Baker_Street_Linux_Server: Tem
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRe
Get:2 http://archive.ubuntu.com/ubuntu jammy InRelease [270
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRele
Get:4 http://archive.ubuntu.com/ubuntu jammy-backports InRe
Get:5 http://security.ubuntu.com/ubuntu jammy-security/mult
Get:6 http://security.ubuntu.com/ubuntu jammy-security/rest
Get:7 http://security.ubuntu.com/ubuntu jammy-security/univ
Get:8 http://security.ubuntu.com/ubuntu jammy-security/main
Get:9 http://archive.ubuntu.com/ubuntu jammy/multiverse amd
```

Next I Identified the packages **telnet** and **rsh-client.** **telnet** and **rsh-client** are considered to be security risks. **Telnet** is considered a risk because it transmits data such as passwords in human-readable form (plaintext) making it insecure. **RSH (Remote Shell)** uses unencrypted communication exposing itself to attacks like MITM (man-in-the-middle attacks).

```
root@Baker_Street_Linux_Server:/# which rsh
/usr/bin/rsh
root@Baker_Street_Linux_Server:/# which telnet
/usr/bin/telnet
root@Baker_Street_Linux_Server:/# dpkg -l | grep telnet
ii  telnet                        0.17-44build1                      amd64        basic telnet client
root@Baker_Street_Linux_Server:/# dpkg -l | grep rsh
ii  rsh-client                    0.17-22                            amd64        client programs for
ii  rsh-server                    0.17-22                            amd64        server program for r
```

Then I installed the packages **ufw**, **lynis**, and **tripwire** by running the command:
sudo apt install -y ufw lynis tripwire

Too verify the packages were installed I used the command:
which <package_name>

To verify with more detail I used the command:
dpkg -l | grep <package_name>

```
root@Baker_Street_Linux_Server:/# which lynis
/usr/sbin/lynis
root@Baker_Street_Linux_Server:/# which ufw
/usr/sbin/ufw
root@Baker_Street_Linux_Server:/# which tripwire
/usr/sbin/tripwire
root@Baker_Street_Linux_Server:/# dpkg -l | grep ufw
ii  ufw                           0.36.1-4ubuntu0.1
root@Baker_Street_Linux_Server:/# dpkg -l | grep tripwire
ii  tripwire                      2.4.3.7-4
root@Baker_Street_Linux_Server:/# dpkg -l | grep lynis
ii  lynis                         3.0.7-1
root@Baker_Street_Linux_Server:/#
```

The hardening features these packages provide are as follows:

UFW (Uncomplicated Firewall)

- Makes firewall management simple.
- Allows necessary connections while at the

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<td>same time blocking unauthorized ones.
Allows firewall rules to be easily configured
Lynis
Security auditing tool for Linux.
Looks for vulnerabilities, misconfigurations, and gaps in security.
Makes suggestions to improve system security
Tripwire
Is an intrusion detection system (IDS)
Will monitor file integrity and alerts when files are suddenly modified.
Can help detect unauthorized modifications to critical system files</td>
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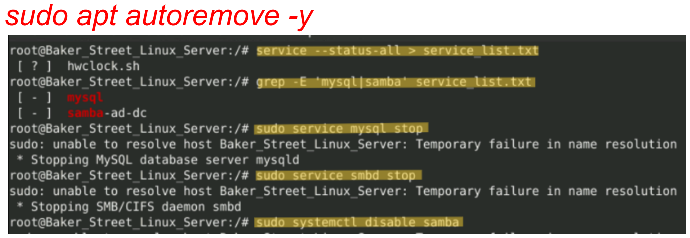
<tr>
<td>☑</td>
<td>Disabling unnecessary services</td>
<td>To start the process of reviewing and disabling unnecessary services, I first ran the command service --status-all > service_list.txt to list all services then output them into the file call service_list.txt.

Next I identified the running services were mysql and samba by using the command:
grep -E 'mysql|samba' service_list.txt.

Then I stopped the services by running the command:
sudo service <service_name> stop

Disabled mysql and samba by using the command:
sudo systemctl disable <service_name>

Next I removed the services running command:
sudo apt remove -y <system_name> <system_name>

Lastly I remove any unused dependencies by using:
sudo apt autoremove -y

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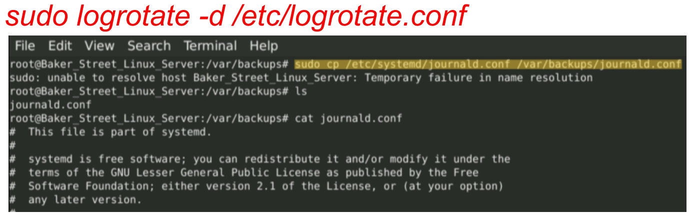
<tr>
<td>☑</td>
<td>Enabling and configuring logging</td>
<td>
Started by opening the journald.conf file for editing using sudo nano /etc/systemd/journald.conf.
Found and updated the following lines:
Storage=persistent
SystemMaxUse=300
Storage=persistent saves logs permanently to your local machine instead of using RAM
SystemMaxUse=300 Restricts the log storage usage to 300MB to prevent excessive disk space usage
Next I edited the logrotate.conf file by running the command sudo nano /etc/logrotate.conf
I then found and update the following lines:
rotate log files weekly
Daily
keep 4 weeks worth of backlogs
Rotate 7
Daily Ensures logs are rotated every day instead of weekly.
Rotate 7 Keeps logs for 7 days before deleting old ones.
Lastly I verified and tested the log rotation changes by running the command:
sudo logrotate -d /etc/logrotate.conf

<pre>
File Edit View Search Terminal Help
root@Baker_Street_Linux_Server:/var/backups# sudo cp /etc/systemd/journald.conf /var/backups/journald.conf
sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in name resolution
root@Baker_Street_Linux_Server:/var/backups# ls
journald.conf
root@Baker_Street_Linux_Server:/var/backups# cat journald.conf
This file is part of systemd.
#
systemd is free software; you can redistribute it and/or modify it under the
terms of the GNU Lesser General Public License as published by the Free
Software Foundation; either version 2.1 of the License, or (at your option)
any later version.
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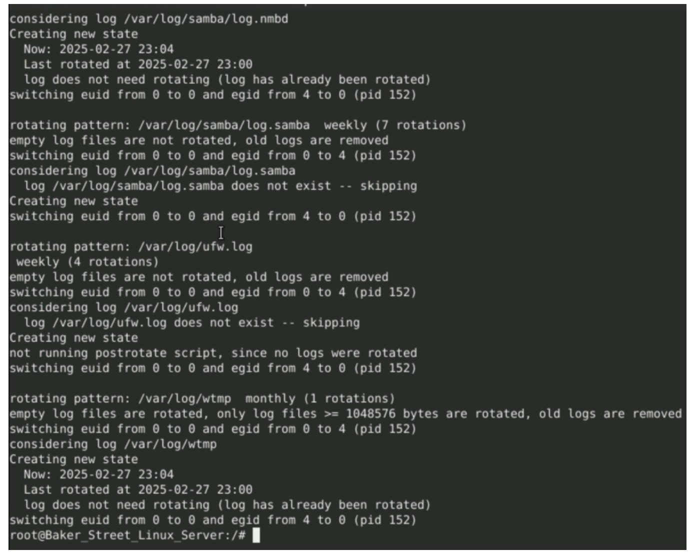
<td>
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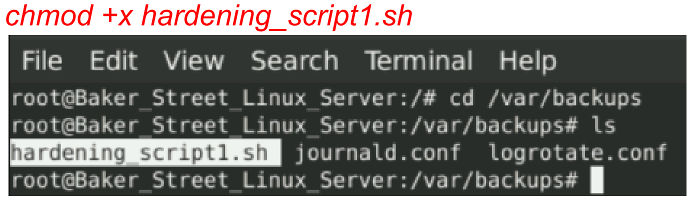
<tr>
<td>☑</td>
<td>Scripts created</td>
<td>I started by creating an output file named bcs_project1.txt. This is where the output of the hardening_script1.sh will be saved.

Then created a backup of the hardening_script1.sh
by running the following command:
sudo cp hardening_script1.sh /var/backups/hardening_script1.sh

Then I proceeded to edit the script by running:
sudo nano hardening_script1.sh

After completing and saving the script I then made the script executable by running the command:
chmod +x hardening_script1.sh


Then I ran the script using the command:</td>
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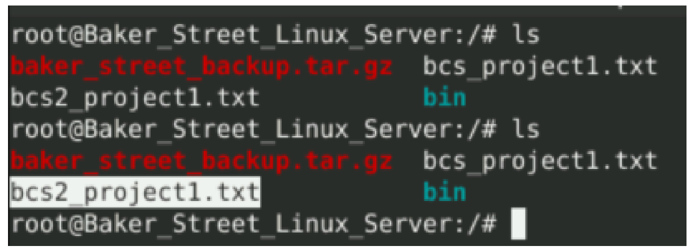
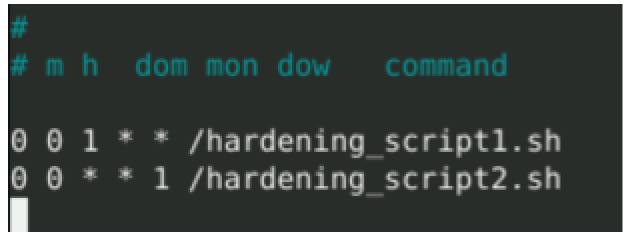
		sudo ./hardening_script1.sh For the second script I started by creating another output file named ***bcs2_project1.txt.*** This is where the output of the **hardening_script2.sh** will be saved.  Then created a backup of the **hardening_script2.sh** by running the following command: *sudo cp hardening_script1.sh /var/backups/hardening_script2.sh* Then I proceeded to edit the script by running: *sudo nano hardening_script2.sh* After completing and saving the script I then made the script executable by running the command: *chmod +x hardening_script2.sh* Then I ran the script using the command: *sudo ./hardening_script2.sh*
☑	Scripts scheduled with cron	To schedule scripts with cron I used the following command to start editing the crontab: *crontab -e* To schedule hardening_script1.sh to run Once a month on the first of the month I inputted: ***0 0 1 * * /hardening_script1.sh*** To schedule hardening_script2.sh to run Once a week every Monday, I inputted: ***0 0 * * 1 /hardening_script2.sh***

Technical Brief Summary Report

Prepared by: Chontele Coleman

The Baker Street Corporation (BSC)

As a security professional I was contacted by **The Baker Street Corporation (BSC)** to harden a Linux server they owned. The BSC Linux server contains sensitive and highly confidential data that they would like to keep secure. Over a three day period I was given the tasks to confirm that their system was indeed properly configured to protect them from security breaches, as well as determine any security issues I came across along the way. If any security issues did arise I was given the authority to make the necessary updates.

Before getting started with my security analyst of the BSC Linux server. It is good practice to record vital information pertaining to the system I will be working on. Such as, host name of the server, OS version, memory info, and uptime information. Last and certainly not least I performed an OS backup. Performing an OS backup can help recover from hardware failure, accidental deletions, virus attacks, or other occurrences.

Day 1: The main takeaway of day 1 is the **principle of least privilege (PoLP)**. According to Wikipedia the definition of **PoLP, requires that in a particular abstraction layer of a computing environment, every module must be able to access only the information and resources that are necessary for its legitimate purpose.** Because employees are your first line of defense and often the most vulnerable, It is important to limit employee access to unnecessary information in case their user accounts are compromised by an attack.

I was given a list of employees that detailed their current status with BSC and performed and executed the following tasks:

- Removed all staff that had been terminated including all home directories and files associated with those staff members.
- Locked all user accounts of staff on temporary leave.
- Unlocked staff members whose status was listed as currently employed.
- Moved all the employees who were in the marketing department to a new group called research. Research group did not previously exist so it had to be created.
- Removed the marketing group because that department closed this year.
- Updated password requirements for all users based on the criteria given by BCS.
- Gave full sudo privileges to the employee specified by BSC and removed all other full privileged employees
- Gave sudo privileges to specified employees to run scripts in a designated location
- Employees in the research group were given sudo privileges only to run a designated script.
- Removed all **read, write, execute (RWX)** permissions from every user's files in their home directory.
- Located employee files with hidden passwords. Removed those files because storing passwords on the server is not only prohibited but poses a security risk to the corporation and the employee.

Day 2: On this day we audited, reviewed, updated packages, disabled unnecessary services, as well as enabled and configured logging. These practices are key to the security and stability of the system. They can also assist and help reduce unrealized exposures confirming only necessary services and packages are running on the system. Keeping these this in good standing greatly reduces the attack surface of the Baker Street Linux Server.

- Configured SSH to not allow the ability to: SSH with empty passwords, SSH with the root user, SSH with any other ports besides Port 22
- Enabled Protocol 2
- Restart the SSH service to set my updates
- Updated the package manager to ensure I was working with the latest version of all packages
- Upgraded all packages to ensure I was working with the latest versions.
- Created a file that contains all installed packages
- Identified telnet and rsh-client as high security risk packages for the following reasons:
  - **telnet** and **rsh-client** are considered to be security risks. **Telnet** is considered a risk because it transmits data such as passwords in human-readable form (plaintext) making it insecure. **RSH**

(Remote Shell) uses unencrypted communication exposing itself to attacks like **MITM (man-in-the-middle attacks).**

- Because of the above mentioned concerns telnet and rsh-client were removed along with all unnecessary dependencies of those packages.
- Added the **ufw, lynis, tripwire.** The benefits of these packages are listed below:

  UFW (Uncomplicated Firewall)

  - Makes firewall management simple.
  - Allows necessary connections while at the same time blocking unauthorized ones.
  - Allows firewall rules to be easily configured

  Lynis

  - Security auditing tool for Linux.
  - Looks for vulnerabilities, misconfigurations, and gaps in security.
  - Makes suggestions to improve system security

  Tripwire

  - Is an intrusion detection system **(IDS)**
  - Will monitor file integrity and alerts when files are suddenly modified.
  - Can help detect unauthorized modifications to critical system files
- Ran command to list out all services and created an output file
- Identified that the services **mysql** and **samba** were running. Stopped, Disabled, and removed those services
- Accessed the journald.conf and proceeded to make edits:
  - **storage=persistent** this setting will save logs locally on the machine.
  - **systemMaxUse300M** configures the maximum disk space the logs can utilize.
- To Prevent logs from taking up too much space it was essential to edit the logrotate.conf file.
  - Changed log rotation from weekly to daily
  - Rotated out logs after 7 days

Day 3: Built out 2 scripts to automate the tasks completed over the last 2 days. The first script will cover day one's tasks. The second script will cover day 2 tasks. The automation of tedious task like backing up files, monitoring system resources, and overseeing user accounts can increase efficiency along with an increase in accuracy and help to simplify tasks.

- After completing the scripts and checking them for errors they were ready to be scheduled to perform at specific times using cron:
    - Script 1 is scheduled to run Once a month on the first of the month.
    - Script 2 is scheduled to run Once a week every Monday.

In conclusion, we learned how to greatly reduce the attack surface of the Baker Street Corporation Linux server by following the model of the **CIA triad (Confidentiality, Integrity, Availability)**. **Confidentiality** safeguards personal and company information. **Integrity** confirms verifiability and defends against unauthorized changes to the system. **Availability** makes sure access to the system information can be done as timely and reliably as possible.

My final thought is all the changes that were made in my project to the BSC Linux Server. There is no such thing as 100% protection from a malicious attack. Recommendations I can give is to educate your staff about what to do with personal information as well as handling sensitive company data. Examples of this are not leaving an unattended laptop open signed into a BSC user account at a coffee shop or airport. Clicking on links inside of emails that look suspicious or from unknown senders. Clicking on links from legitimate looking emails with misspellings. My last recommendation to the BSC is to have a contingency plan in case a cyberattack or data breach occurs.